AERODYNAMIC FORENSIC ENGINEERING REPORT

Produced by the Wind_Navigator Data-as-a-Service Engine

REPORT REFERENCE	WN-FR-492C8206
DATE OF ANALYSIS	2026-05-17 14:41:36 UTC
LOG SOURCE FILE	ARDUPILOT_00000135.BIN
INCIDENT LOCATION	Extracted from log: Lat 12.4380, Lon 77.3167
FLIGHT DURATION	1619 seconds
PHYSICS ENGINE	D2Q9 LBM (Integer Remainder Vault v1.0)

1. Executive Determination

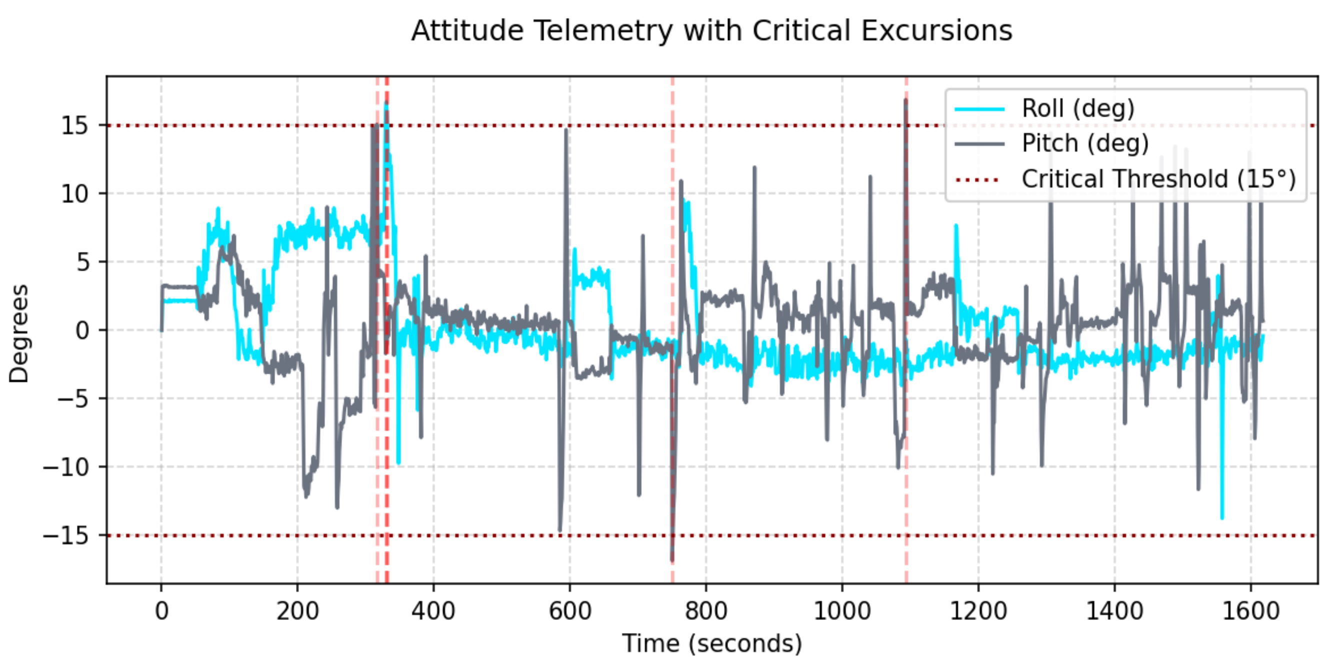
DETERMINATION: AERODYNAMIC FORCES EXONERATED

Mathematical analysis of 18 ambient wind scenarios found NO environmental configuration capable of producing the observed attitude excursions at the recorded crash coordinates. The localized structural geometry does not support vortex formation at this location.

Conclusion: The physics engine has isolated the failure cause to mechanical origin, electrical origin (e.g., ESC desync, motor failure), or pilot error. The ambient environment is mathematically cleared of contributing to this crash.

1.1 Flight Data & Attitude Excursions

The following graph plots the drone's recorded Roll and Pitch angles over the duration of the flight. The dashed red lines indicate timestamps where the flight controller recorded an uncommanded attitude excursion exceeding 15 degrees, leading to loss of control.



2. Forensic Methodology

Standard post-crash investigations suffer from a critical lack of atmospheric data. Black box logs record the drone's reaction, but they do not record the wind itself. Furthermore, urban environments create micro-vortices (building corner shears) that do not appear on standard meteorological reports.

To bypass this limitation, Wind_Navigator employs a **Wind-Vector Back-Propagation Engine**:

- 1. **Geometry Ingestion:** The exact GPS flight path is extracted from the .BIN log.
- 2. **Structural Mapping:** Physical building footprints surrounding the flight path are downloaded from OpenStreetMap and rasterized.
- 3. **Fluid Dynamics Sweep:** A pure-integer Lattice Boltzmann Method (D2Q9) physics engine simulates 18 distinct ambient wind directions (0° to 340°).
- 4. **Cross-Referencing:** The resulting turbulence maps are overlaid onto the crash coordinates to check for mathematical correlation.

2.1 Back-Propagation Sweep Results

Wind Heading	True Positives (Matched Anomalies)	Conclusion
0°	0	No Correlation
20°	0	No Correlation
40°	0	No Correlation
60°	0	No Correlation
80°	0	No Correlation
100°	0	No Correlation
120°	0	No Correlation
140°	0	No Correlation
160°	0	No Correlation
180°	0	No Correlation

200°	0	No Correlation
220°	0	No Correlation
240°	0	No Correlation
260°	0	No Correlation
280°	0	No Correlation
300°	0	No Correlation
320°	0	No Correlation
340°	0	No Correlation

3. Certificate of Determinism

Standard computational fluid dynamics (CFD) engines use IEEE 754 floating-point mathematics. Due to floating-point truncation, running the same simulation on an Intel CPU versus an NVIDIA GPU or an ARM processor will produce slightly different numerical outputs. In legal contexts, this non-reproducibility can invalidate the forensic analysis.

Wind_Navigator operates exclusively on **Rational Trigonometry and Integer Arithmetic** utilizing a proprietary Remainder Vault to strictly enforce mass conservation. Because no floating-point mathematics are executed during the physics step, the engine is 100% bit-deterministic.

Running this exact flight log through the engine on any hardware architecture will produce the exact same SHA-256 binary output signature.

Physics Engine SHA-256 Checksum:

```
9d6e91ec3c940a34fd0ba3109d7a928178039075de72fe7506d1256af65b6993
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End of Report